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To the Editors-in-Chief of *Computers and Electronics in Agriculture*

Dear Editors,

I submit the manuscript **"Differentiable simulation for agrivoltaic tracker control under a yield floor: what learning adds and what shade-response uncertainty costs"** for consideration as an original research article in *Computers and Electronics in Agriculture*.

What is new. The manuscript develops, rather than applies, tools for control and design of agrivoltaic systems, which put crops and photovoltaic modules on the same land under a yield-retention floor required by regulators or farmers. It contributes (i) a differentiable simulator that couples tracker geometry, photovoltaic output, light and microclimate under the modules, soil water and crop growth, so that yield and electricity are differentiable with respect to design and control variables; (ii) a neural controller trained through whole seasons by back-propagation, instead of reinforcement learning with black-box simulators, which makes 300 gradient steps sufficient; (iii) an evaluation protocol in which the array density of every site is chosen from its own history only and performance is measured on geographically held-out sites and years, for 400 crop–site pairs, five crops and four climates; and (iv) methods for design under uncertainty about the crop response to shade, namely an ensemble of responses obtained by gradient calibration through the simulator, margins calibrated by conformal risk control, and a learned design network.

Fit to the journal. The work lies in the journal's areas of artificial intelligence, control systems and simulation modelling for agriculture, and the simulator, the controller and the design methods are investigator-developed tools, not off-the-shelf components.

Main results, including negative ones. Light-sharing rules allow 2.1 to 2.9 times the module density of static single-axis arrays at the same mean retention, but the floor is missed in 24–31% of seasons. The learned controller exceeds a written phenology rule by 0.044 in land-equivalent ratio but an optimised window by only 0.007, with no detectable difference at matched retention. Shade curves learned from 428 field observations are no better than proportional loss for the staple crops, and the simulator is milder than the data for maize. A density chosen under the mean response leaves 70% of pairs at the floor when another plausible response holds, against 87% for an oracle design; conformal margins attain a stated compliance inside the ensemble of responses, a learned design network does worse than a simple margin, and only large margins are safe outside the ensemble. The manuscript reports these null and negative findings plainly.

Limits. The evaluation is by simulation on public data (SPAM, GGCMI, NASA POWER and a published shade meta-analysis), without field experiments under trackers; I state this in the abstract and in the limitations, and I identify field validation under trackers, in particular for maize, as the next step. The economic and scale estimates are secondary and are in the supplementary material.

Declarations. The work is original, has not been published, and is not under consideration elsewhere. The declaration of competing interests is provided in the submission system. The

data are public; the code, the site list and the result tables are in a public repository, and a DOI-registered archive will be cited before publication. The submission includes the manuscript, highlights, a graphical abstract and supplementary material.

Thank you for considering the manuscript.

Sincerely,

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